Why surds are important? For calculation purposes we assume approximate value as $\sqrt{2} = 1.414, \sqrt{3} = 1.732$ and so on.

$$\left(\sqrt{2}\right)^2 = \left(1.414\right)^2 = 1.99936 \neq 2 \; ; \quad \left(\sqrt{3}\right)^2 = \left(1.732\right)^2 = 3.999824 \neq 3$$

Hence, we observe that $\sqrt{2}$ and $\sqrt{3}$ represent the more accurate and precise values than their assumed values. Engineers and scientists need more accurate values while constructing the bridges and for architectural works. Thus it becomes essential to learn surds.



Progress Check

1. Which is the odd one out? Justify your answer.
   (i) $\sqrt{36}$, $\sqrt{\frac{50}{98}}$, $\sqrt{1}$, $\sqrt{1.44}$, $\sqrt[5]{32}$, $\sqrt{120}$
   (ii) $\sqrt{7}$, $\sqrt{48}$, $\sqrt[3]{36}$, $\sqrt{5}+\sqrt{3}$, $\sqrt{1.21}$, $\sqrt{\frac{1}{10}}$
2. Are all surds irrational numbers? - Discuss with your answer.
3. Are all irrational numbers surds? Verify with some examples.

2.6.1 Order of a Surd

The order of a surd is the index of the root to be extracted. The order of the surd $\sqrt[n]{a}$ is n. What is the order of $\sqrt[5]{99}$? It is 5.

Surds can be classified in different ways:

(i) Surds of same order : Surds of same order are surds for which the index of the root to be extracted is same. (They are also called equiradical surds).

For example, $\sqrt{x}$, $a^{\frac{3}{2}}$, $\sqrt[2]{m}$ are all 2nd order (called quadratic) surds .

$\sqrt[3]{5}$, $\sqrt[3]{(x-2)}$, $(ab)^{\frac{1}{3}}$ are all 3rd order (called cubic) surds.

$\sqrt{3}$, $\sqrt[3]{10}$, $\sqrt[4]{6}$ and $8^{\frac{2}{5}}$ are surds of different order.

(ii) Simplest form of a surd : A surd is said to be in simplest form, when it is expressed as the product of a rational factor and an irrational factor. In this form the surd has

(a) the smallest possible index of the radical sign.

(b) no fraction under the radical sign.

(c) no factor is of the form a^n, where a is a positive integer under index n.

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